ENGINEERING PORTFOLIO 2023 — 2026

Vedant Nadoda

Mechanical Engineer

ADDITIVE MANUFACTURING · MECHATRONICS · MANUFACTURING

Hands-on engineer who designs, builds, tests and troubleshoots physical systems — from industrial large-format 3D printers to closed-loop mechatronic prototypes and metal-AM test specimens.

EDUCATION

M.S. Mechanical Engineering
New York University · Dec 2026

CONTACT

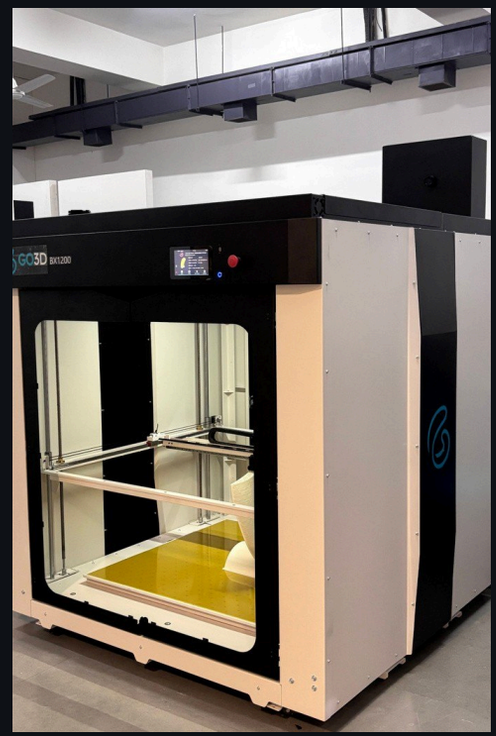
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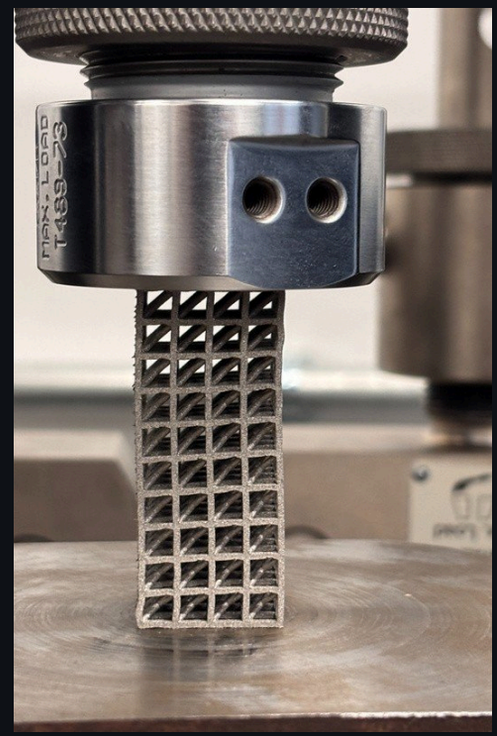
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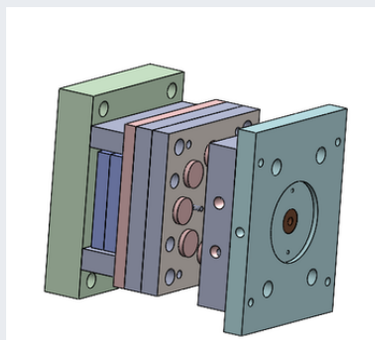
BASED IN

New York City Metro Area
github.com/Vedant030301









P01 Large-format FDM printer build & commissioning

P03 DMLS 316L lattice under UTM compression

P02 Closed-loop peristaltic pump, STM32 control

P05 Eight-cavity injection mold, SolidWorks

ENGINEERING PROFILE

Design it. Build it. Test it.
Find out why it failed — and fix it.

Mechanical engineer with 2+ years of hands-on experience building industrial FDM printers, prototyping mechanisms and testing materials across metal (DMLS/LPBF, WAAM) and polymer (FDM, SLA) platforms. My work spans the full loop: CAD and DFM/DFA up front; assembly, calibration and commissioning on the floor; mechanical testing and root-cause analysis when results don't match the spec.

EDUCATION & CERTIFICATION

M.S. Mechanical Engineering — Jan 2025 – Dec 2026
New York University

B.Tech Mechanical Engineering — 2020 – 2023
Indus University, India

Advance Certification — Machining Technology & Plastic Engineering — Jun 2024
International Automobile Centre of Excellence (JV of Govt. of Gujarat & Maruti Suzuki) · Graded "Competent"

TARGET ROLES — FULL-TIME FROM DEC 2026

Mechanical Engineering · Additive Manufacturing · Manufacturing Engineering · Product Development · R&D / Prototyping · Mechatronics · Applications Engineering

AT A GLANCE

67% ↓
Printer assembly lead time
3 days → 1 day · GO3D (P01)

7
Client installations supported
Setup, calibration, acceptance (P01)

1200 mm
Largest FDM build volume built
Also 500 mm and 300 mm (P01)

150+
Students supported in labs
SEM, XRD, tensile, hardness · NYU

SELECTED WORK

P01	**Industrial Large-Format FDM Printers** — additive mfg	03
P02	**Closed-Loop Peristaltic Pump** — mechatronics	05
P03	**DMLS 316L Stainless-Steel Lattices** — metal AM	07
P04	**Hydraulic Lathe Cross-Slide** — mechanism design	09
P05	**Injection Mold Design** — plastics & tooling	10
P06	**CNC / VMC Machining** — G-code & metrology	11
P07	**Rapid Build & Simulation** — plotter · solver	12
–	**Experience & Capabilities** — skills mapped to projects	13

Design	Build	Test	Analyze	Debug	Improve
P01 P04 P05	P01 P02 P07	P02 P03	P03 P07	P01 P02	P01

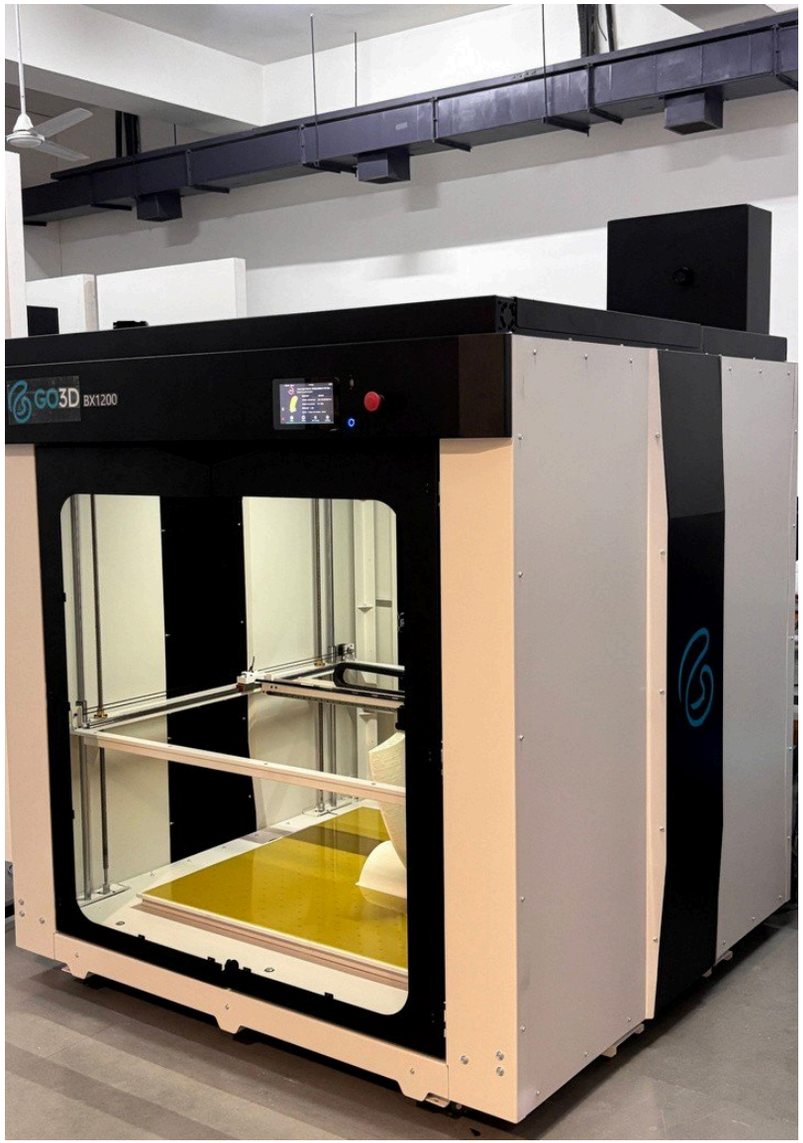


FIG 1.1 Enclosed large-format FDM platform — the class of machine assembled, calibrated and quality-validated before customer handover.

BUILD VOLUMES DELIVERED TO CLIENT SPEC

300 mm	500 mm	1200 mm

P01 · INDUSTRY · GO3D INNOVATION LLP

Industrial Large-Format FDM Printers — Build, Calibration & Commissioning

Built, calibrated and commissioned custom FDM printers up to 1200 mm build volume, then carried them through customer installation, operator training and support.

ROLE	DURATION	TOOLS
Production Support Engineer Mech. Engineering Intern (prior)	Nov 2023 – Dec 2024 Gandhinagar, India	SolidWorks · Cura · GD&T Lean · Quality inspection

01 PROBLEM / OBJECTIVE

Clients ordered printers to custom specifications at **1200, 500 and 300 mm** build volumes. Each machine had to be assembled, calibrated and validated to spec, then installed and supported on site — and assembly took **3 days per machine**.

02 ENGINEERING APPROACH — THE LIFECYCLE I SUPPORTED

FACTORY	01 **Assemble** mechanical build, modifications	02 **Calibrate** hardware–software integration	03 **Test & QA** print-quality validation
CUSTOMER	04 **Install** site setup, final acceptance	05 **Train** operators, SOPs, maintenance	06 **Support** + custom part production

Design input — SolidWorks
components · fixtures · cable mgmt · mounts

Feedback — root-cause analysis
mechanical & process deviations traced

Process improvement — Lean · 5S · Kaizen · standardized workflows
assembly, inventory, machine organization, BOM accuracy

3 days → 1 day

03 DESIGN & DEVELOPMENT

- **Designed and validated** custom machine components, cable-management systems, fixtures and mounting hardware in SolidWorks, applying DFM/DFA and GD&T.
- **Designed and ran** iterative DOE-style test plans across FDM and resin print parameters to find process limits, quantify accuracy, and improve calibration, slicing and post-processing (intern phase).

04 BUILD / IMPLEMENTATION

- **Built, assembled and commissioned** printers hands-on — assembly, disassembly, prototype modifications, calibration and testing — to client specifications.
- **Supported on-site installations:** equipment setup, site requirements, calibration, troubleshooting and final acceptance.

05 TESTING & ITERATION

- **Calibrated and validated** print quality; integrated hardware and software.
- **Performed root-cause analysis** on mechanical and process deviations.
- **Wrote SOPs** and trained customers and operators on operation, maintenance, calibration and troubleshooting.



FIG 1.2 Production floor — large-format frames mid-assembly beside smaller printers.



FIG 1.3 Fine relief, thin walls and tall features — geometry that exposes calibration errors.

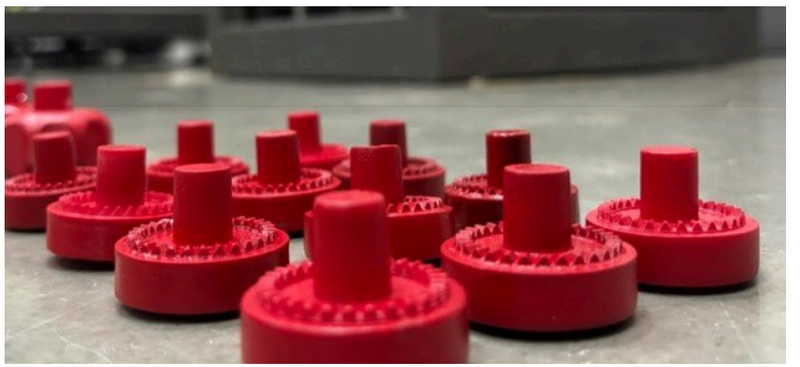


FIG 1.4 Customer-specific components printed in batch — repeatability across a run is the quality target.

06 RESULTS

67% ↓
Assembly lead time, 3 → 1 day
Standardized Lean workflow (team effort)

98%
Dimensional accuracy
Root-cause & process work contributed

7
Client installations supported
Setup through final acceptance

3
Industry exhibitions
Live demos & client engagement

07 TAKEAWAY

Production-floor engineering: taking a machine from parts to a calibrated, validated product; tracing the root cause when prints go out of spec; and improving the process that builds the machine — not just the machine itself.

P02 · NEW YORK UNIVERSITY

Closed-Loop Peristaltic Pump with Water-Level Control

Developed an automated peristaltic pumping system for electronically controlled fluid transfer and water-level regulation.

ROLE	DURATION	TOOLS
Control logic, electrical architecture, system integration	Jan – May 2026	STM32 H5 Nucleo · TMC2209 · Stepper · LCD · Keypad · 3D printing · Laser cutting

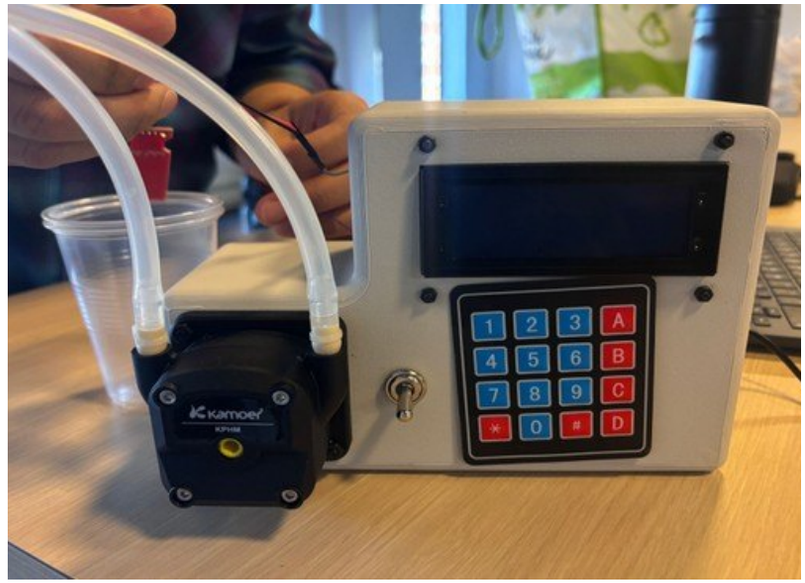


FIG 2.1 Working prototype: stepper-driven pump head, enclosure with LCD and 4×4 keypad, emergency switch, tubing to the reservoir.

01 PROBLEM / OBJECTIVE

Hold a target water level without manual pump operation. The system has to **sense** the level, **decide** when to pump, **drive** a peristaltic pump, and let a user **monitor and set** it — all in one self-contained unit.

02 ENGINEERING APPROACH — CLOSED-LOOP SYSTEM ARCHITECTURE

USER: **Keypad** IN · **LCD** OUT

SENSE: ① SENSOR — **Water-level sensor** — measures reservoir level

COMPUTE: ② MICROCONTROLLER — **STM32 H5 Nucleo** — control logic: pump on / off

DRIVE: ③ MOTOR DRIVER — **TMC2209** — step / direction to motor

ACTUATE: ④ ACTUATOR — **Stepper motor** — turns the pump rollers

↓

⑤ PUMP — **Peristaltic pump** — rollers squeeze the tubing

← ⑥ PROCESS — **Fluid transfer** — through tubing to reservoir

← ⑦ STATE — **Water level changes** — sensed again by ①

← **water-level feedback** CLOSES THE LOOP ↑

POWER & SAFETY **24 V DC supply** → **Emergency switch** → **DC-DC step-down** → separate voltage levels for motor, display and control

03 DESIGN & DEVELOPMENT

CONTROL	Closed-loop logic on the STM32: water-level feedback decides when the pump actuates.
ACTUATION	Drove a stepper-powered peristaltic pump through a TMC2209 driver. Only the tubing touches the fluid.
POWER	Electrical architecture around a 24 V supply with DC-DC conversion for each subsystem; added an emergency switch.
INTERFACE	LCD for system status and a keypad for user input.

05 TESTING & ITERATION

Verified end-to-end operation: level feedback triggered pump actuation automatically, and the LCD/keypad interface worked as intended.

LIMITATION FOUND → NEXT ITERATION

The water-level sensor showed **corrosion and reliability issues** in use. A v2 should prioritize level sensing suited to long-term water exposure — the weakest link in the loop.

04 BUILD — WIRING ARCHITECTURE

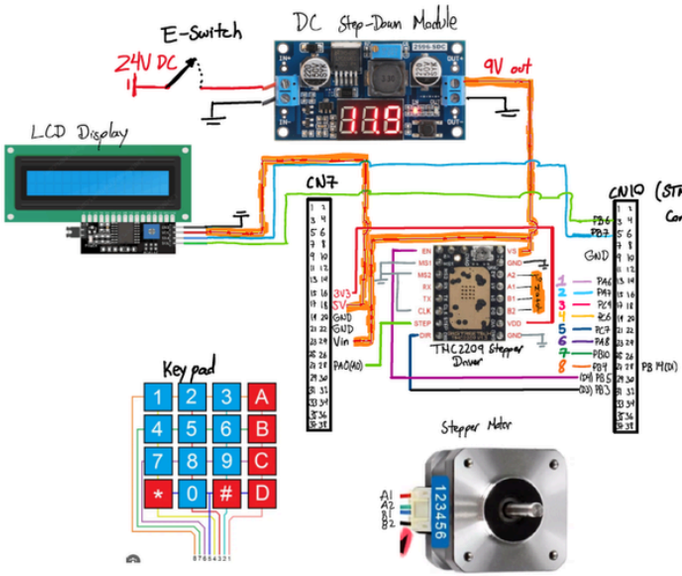


FIG 2.2 Wiring map from supply to actuator: 24 V input, E-switch, DC step-down module, TMC2209, stepper, LCD and 4×4 keypad, pinned to the STM32 H5 Morpho connectors (CN7 / CN10).

- **Integrated** tubing, pump, level sensor, electronics and reservoirs into one physical prototype.
- **Fabricated** structural components with 3D printing and laser cutting.

06 RESULTS

Closed-loop regulation
Water level held automatically through pump actuation

Autonomous transfer
Driven by real-time level feedback — no manual pumping

Working HMI
LCD status display and keypad input demonstrated

Integrated prototype
Mechanical, electrical, sensing, code and UI in one unit

07 TAKEAWAY

Mechatronic system integration — sensor, microcontroller, motor driver, power electronics and mechanical hardware working as one closed loop — plus the judgment to identify the least reliable component and target it for the next iteration.

DMLS-Printed 316L Stainless-Steel Lattices — Tension & Compression Characterization

Designed, printed and tested 316L lattice structures to characterize stiffness, strength and failure behavior for lightweight, energy-absorbing applications.

DURATION	PROCESS	TESTING	ANALYSIS
Sep – Dec 2025	DMLS metal powder-bed fusion	Instron UTM · tension & compression	Force–displacement → stress–strain



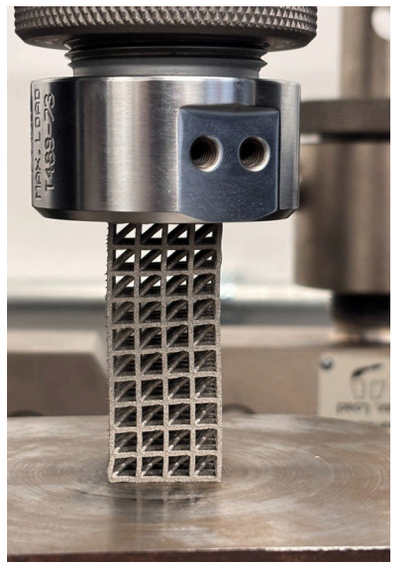
A Build orientation with supports



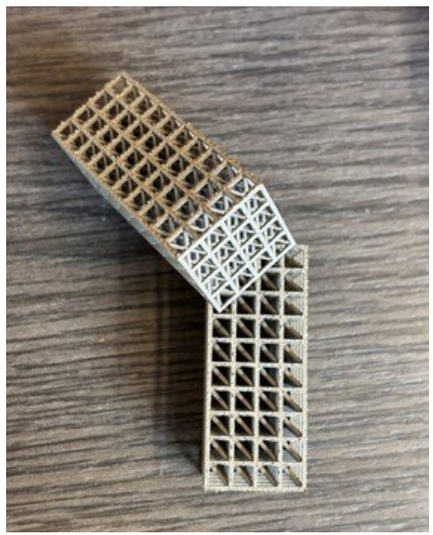
B Tensile specimen



C Tensile test



D Compression between platens



E Deformed lattice — failure mode is the result

01 PROBLEM / OBJECTIVE

Metal AM lattices can make light, energy-absorbing structures — but stiffness and strength depend on geometry and print quality. Goal: measure **stiffness, yield and ultimate strength** in tension and compression, and identify **how the lattices fail**.

02 ENGINEERING APPROACH — TEST WORKFLOW

01 PREP	02 PRINT	03 TEST	04 REDUCE	05 INTERPRET
Build setup orientation, scan, supports	**DMLS · 316L** tensile + compression specimens	**Instron UTM** 5 samples per load case	**Stress–strain** from raw force–displacement	**Failure modes** tearing vs. buckling

03 DESIGN & DEVELOPMENT

Process decisions for repeatable specimens

ORIENTATION	**45° (+X) and 15° (+Z)**
SCAN	Strategic scan patterns
SUPPORTS	Stabilized support structures
MATERIAL	SS316L powder, DMLS

05 TESTING & ITERATION

- **Tested** 5 tensile and 5 compression specimens on an Instron UTM.
- **Converted** raw force–displacement data into engineering stress–strain curves to compare samples.
- **Characterized failure:** strut tearing in tension; progressive buckling in compression.
- Compressive modulus varied (0.44–1.50 GPa) — attributed to lattice buckling, while strength stayed consistent.

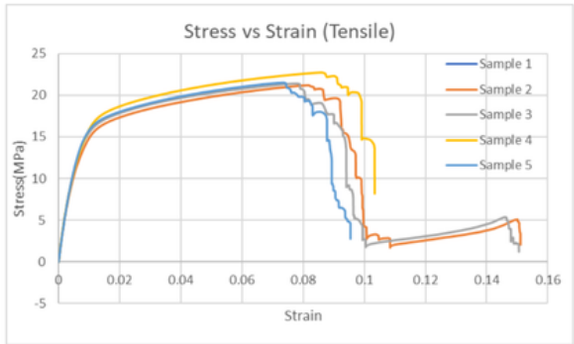


FIG 3.1 Tension, 5 samples: tight clustering to peak, then strut tearing drives the drop.

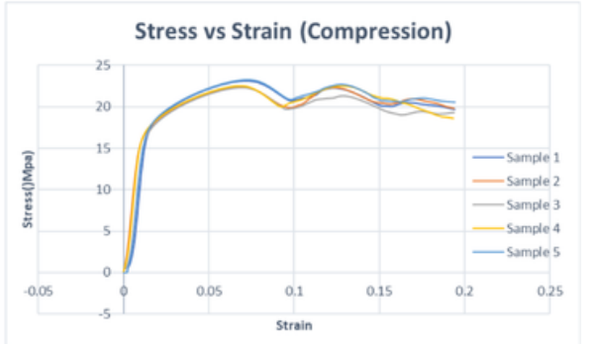


FIG 3.2 Compression, 5 samples: stress plateaus instead of dropping — progressive, non-brittle buckling.

06 RESULTS

LOAD CASE	STRENGTH	MODULUS	FAILURE MODE
Tension	**21–23** MPa UTS	**2.4–2.5** GPa	Strut tearing
Compression	**22–23** MPa	**0.44–1.50** GPa	Progressive buckling

Stable, repeatable tensile properties and a steady compressive strength confirm a **controlled, non-brittle failure profile** suited to energy-absorbing applications.

07 TAKEAWAY

The full metal-AM qualification loop at small scale: build-process decisions, standardized mechanical testing, data reduction and failure-mode interpretation — the same workflow used to decide whether an AM part is fit for service.

Hydraulic Cross-Slide Retrofit for a Conventional Lathe

Designed and prototyped a semi-automated hydraulic cross-slide to replace a manual lathe mechanism — a low-cost alternative to CNC retrofitting.

DATE	TOOLS / SYSTEM
2023	Hydraulic cylinder · power pack · PLC control · CAD drawings

01 PROBLEM

Manual cross-feed demands constant operator effort and adds vibration and inconsistency to the cut.

02 APPROACH

Replace the handwheel cross-slide with a **hydraulic cylinder and power pack**, mounted through a **custom plate**, with PLC-based control.

03 DESIGN

Designed and integrated the cylinder, power pack and custom mounting plate, applying hydraulic principles.

05 TESTING

Tested facing, taper turning and grooving operations on the prototype.

06 RESULTS

~70%

Prototype success on trial operations

- Smoother cross-feed motion
- Reduced vibration and operator fatigue
- Improved surface finish
- A low-cost alternative to CNC retrofitting

07 TAKEAWAY

Retrofitting a real machine: integrating fluid power, a custom mount and control onto existing equipment, then validating on actual cutting operations.

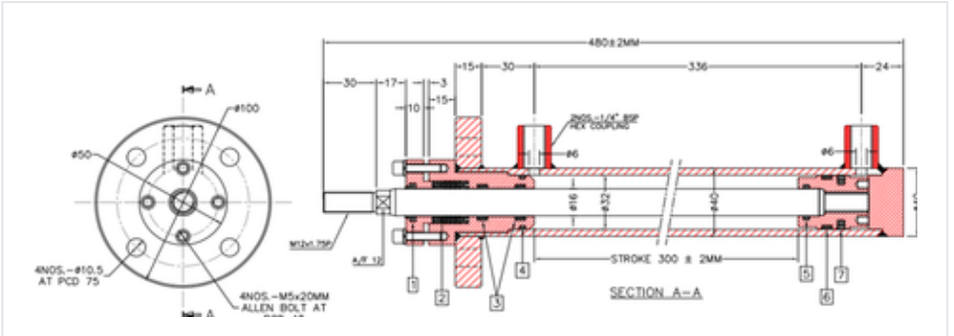


FIG 4.1 Cylinder end view and section A–A: 300 ± 2 mm stroke, 480 ± 2 mm overall length, port locations defined.



FIG 4.2 Cylinder driving the cross-slide

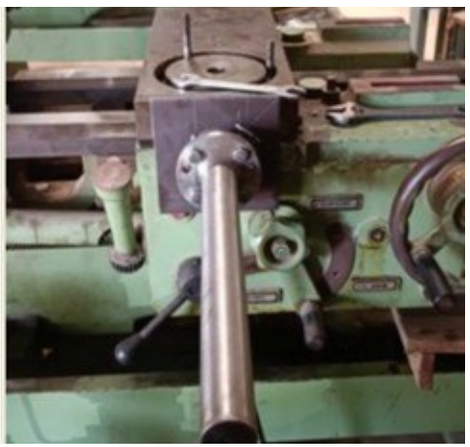


FIG 4.3 Cylinder mounted on the lathe

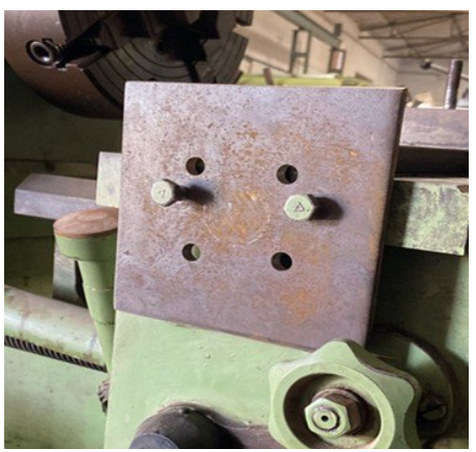


FIG 4.4 Custom mounting plate installed



FIG 4.5 Fabricated cylindrical component

Injection Mold Design — Curtain Ring

Designed a complete injection mold in SolidWorks and gained hands-on time on an industrial injection molding machine.

DATE	TOOLS
2023 – 24	SolidWorks · FANUC Roboshot α-S100iA

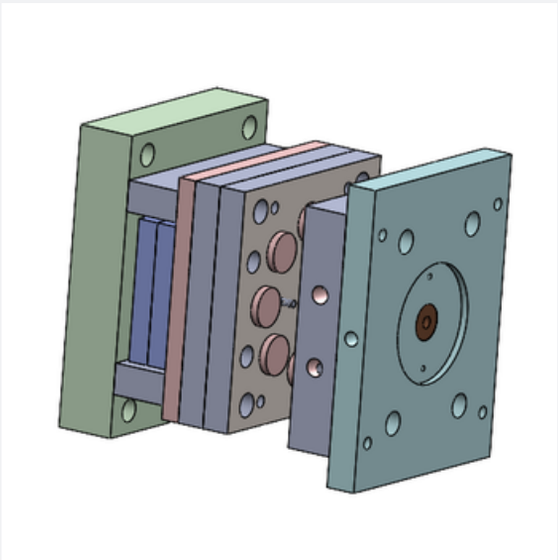


FIG 5.1 Exploded mold assembly — plate stack, core and cavity inserts, guide elements and fasteners designed as one system.

CREDENTIAL

Advance Certification Programme — Machining Technology & Plastic Engineering

International Automobile Centre of Excellence, Gandhinagar (JV of Govt. of Gujarat & Maruti Suzuki)

GRADE · JUN 2024

Competent

01 OBJECTIVE

Design a production-style mold for a curtain ring — every functional element a real tool needs, not just the part cavity.

03 MOLD ANATOMY DESIGNED

CORE & CAVITY · VENTING · EJECTION SYSTEM · COOLING CHANNELS · GUIDE PILLARS · BUSHES · FIXTURES · FASTENERS

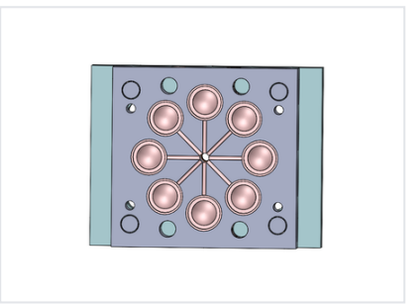


FIG 5.2 Eight cavities fed by a radial runner from the sprue

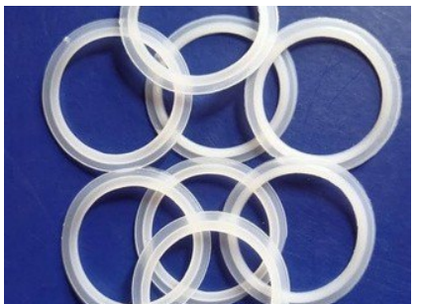


FIG 5.3 Curtain rings — the part the mold produces



FIG 5.4 FANUC α-S100iA

04 HANDS-ON

Ran the FANUC Roboshot α-S100iA and hopper-dryer systems; studied plastic materials, part design, mold anatomy and mold-flow analysis in the program curriculum.

07 TAKEAWAY

Designing tooling forces DFM thinking: feeding, cooling and ejecting the part matter as much as its shape.

CNC / VMC Machining — From Drawing to Part

Designed parts, defined toolpaths, hand-programmed G-code and machined plastic, aluminum and copper-alloy components on FANUC-controlled CNC and VMC machines.

01 DESIGN	02 PLAN	03 PROGRAM	04 MACHINE	05 INSPECT
Part drawing	**Toolpaths & work-holding**	**Manual G-code**	**FANUC CNC / VMC**	**Calipers · micrometers**

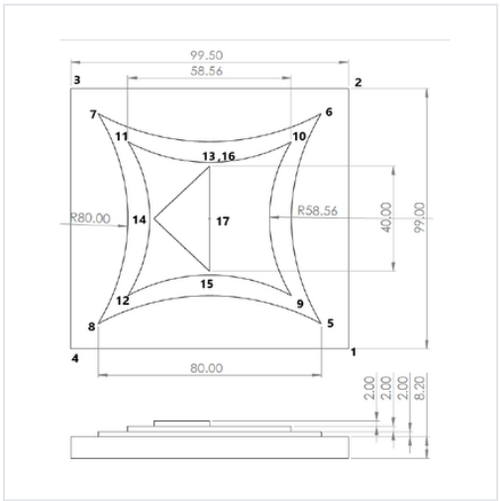


FIG 6.1 Dimensioned drawing with numbered profile points — the reference for toolpath coordinates.

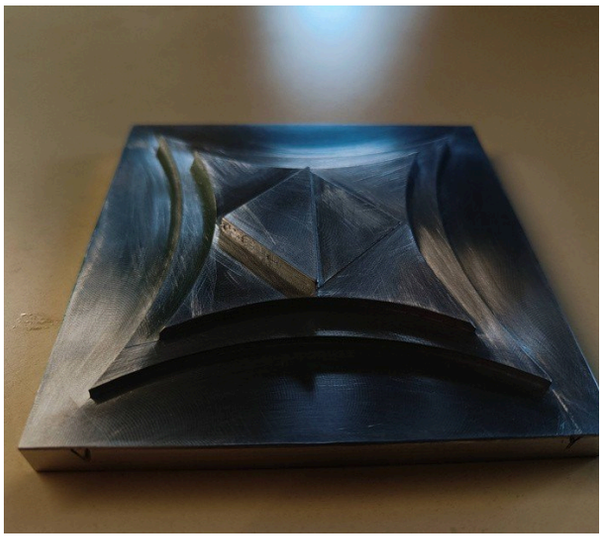


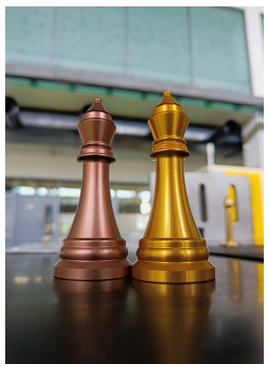
FIG 6.2 The machined part: concave-arc profile, stepped pockets and a raised triangular feature.

03 WHAT I DID

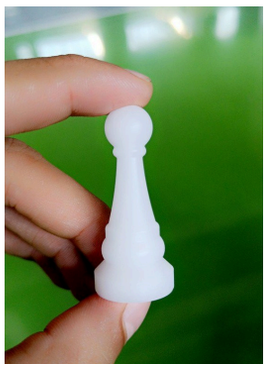
- **Programmed** G-code by hand — linear and circular interpolation, canned cycles.
- **Applied** cutting parameters, machining process sequence and work-holding principles.
- **Machined** components from raw plastic, aluminum and copper-alloy stock, producing 3 validation samples per operation.
- **Inspected** parts with vernier calipers, micrometers and dial indicators; trained in GD&T.



6.3 Raw stock



6.4 Turned profiles



6.5 Turned profile



6.6 Multi-diameter

07 TAKEAWAY

Knowing how a part is actually cut — toolpaths, fixturing, tolerances — makes every design I draw easier and cheaper to manufacture.

P07A · NYU MAKERSPACE · 2025

Mini CNC Plotter — 12-Hour Build

Designed and assembled a functional CNC plotter from scrap materials in under 12 hours.

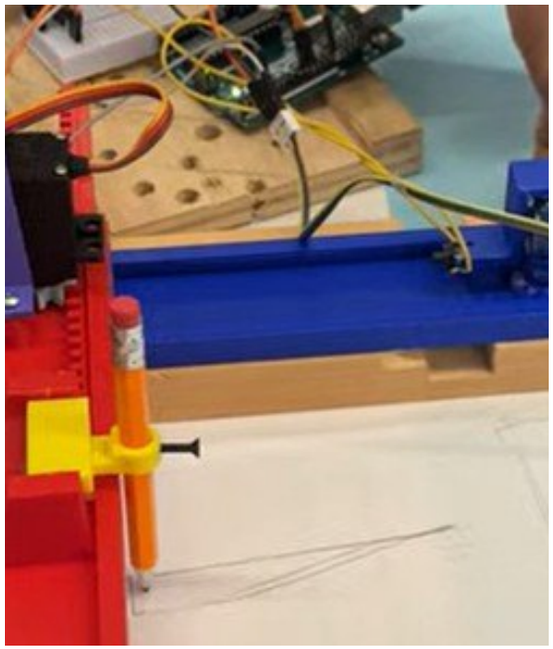


FIG 7.1 Servo axes on a 3D-printed rack

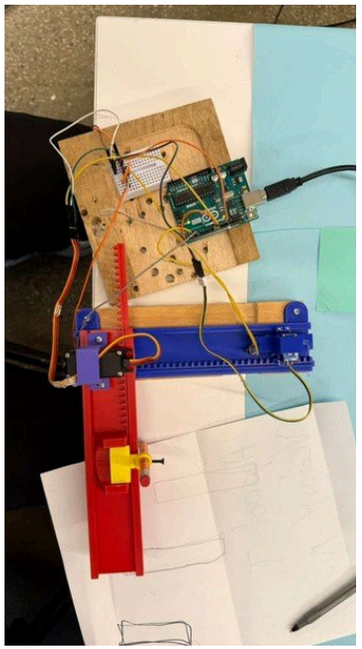


FIG 7.2 Arduino on scrap wood

1st
Place, NYU Makerspace Scrapathon

<12h
Concept to working machine

$200
Grant for future projects

How: Arduino control, servo motors and a 3D-printed structure, built from scrap materials under a hard time limit.

TAKEAWAY

Rapid prototyping under constraint: choose simple mechanisms, print what's missing, and get to a working machine fast.

P07B · NEW YORK UNIVERSITY · SEP – DEC 2025

Couette Flow — Thermal-Fluid Solver

A Python finite-difference solver for coupled Navier–Stokes and energy equations, from incompressible to supersonic regimes.

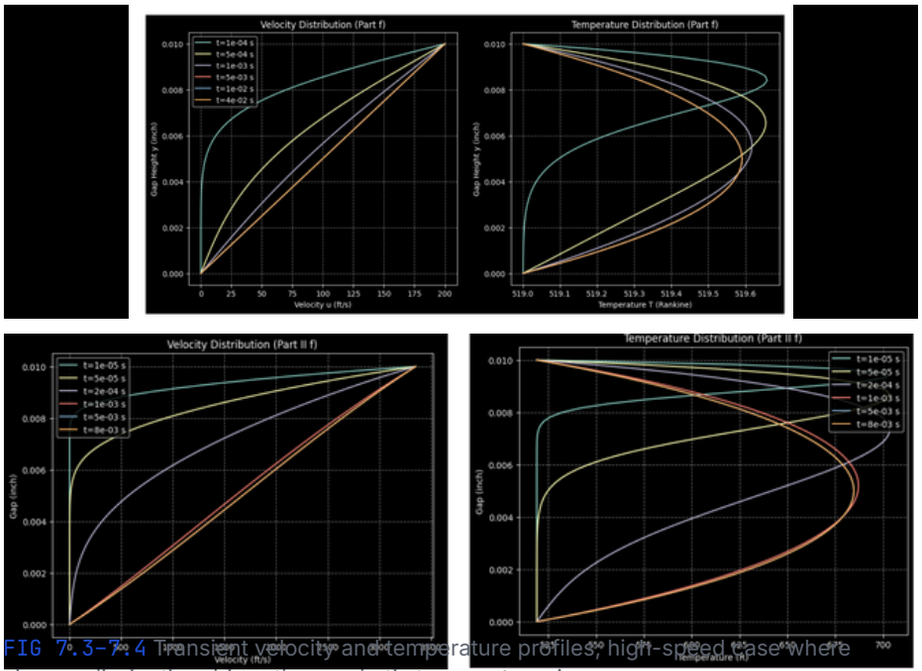


FIG 7.3–7.4 Transient velocity and temperature profiles, high-speed case where viscous dissipation drives the quadratic temperature rise.

521.316 °R
Adiabatic wall temp. at Ma ≈ 3.0

1.30×10^{-4}
Skin-friction coefficient C_f

−3.74
Viscous heat flux, BTU/s·ft²

Verified solution accuracy with grid-independence and time-step sensitivity studies.

TAKEAWAY

Analysis to back up hardware: building and verifying my own numerical model rather than trusting a black box.

EXPERIENCE

Research Assistant — Computational Materials
JUN 2026 – NOW

New York University

MD simulation of melting in Au, Al and Cu (LAMMPS / MedeA), validated against experiment; automated HPC workflows; mentor 2 undergraduate researchers.

Course Assistant — Materials Science Lab
SEP 2025 – NOW

New York University

Lab procedures and training for 150+ students (30+ per session) across XRD, SEM, tensile, hardness, metallography, heat treatment and laser cutting. NYU Lab Safety Certified.

Production Support Engineer
FEB – DEC 2024

GO3D Innovation LLP, Gandhinagar, India · see P01

Mechanical Engineering Intern
NOV 2023 – FEB 2024

GO3D Innovation LLP · DOE print testing, exhibitions, 5S / Kaizen · see P01

Advance Certification Trainee
2023 – 2024

IACE Gandhinagar · Machining Technology & Plastic Engineering · see P05–P06

Mechanical Engineering Intern — Quality
JUN – JUL 2021

Sunshine Enterprise, Ahmedabad · Dimensional verification with calipers, micrometers and gauges against ISO and customer requirements.

SOFTWARE & PLATFORMS

SOLIDWORKS · CREO · AUTOCAD · CAD/CAM · MATERIALISE MAGICS · CURA · PYTHON · LAMMPS / MEDEA · PEEL 3D SCANNER · LPBF / DMLS · WAAM · FDM · SLA



Part models built during SolidWorks training at IACE.

CAPABILITIES — MAPPED TO EVIDENCE

CAPABILITY	P01	P02	P03	P04	P05	P06	P07	NYU
Mechanical design & CAD (SolidWorks)	●	–	●	●	●	●	●	–
Engineering drawings & GD&T	●	–	–	●	–	●	–	–
DFM / DFA	●	–	–	–	–	–	–	–
Polymer AM — FDM / resin	●	●	–	–	–	–	●	–
Metal AM — DMLS build setup	–	–	●	–	–	–	–	–
Prototyping & hands-on assembly	●	●	–	●	–	–	●	–
Mechanical testing (UTM)	–	–	●	–	–	–	–	●
Materials characterization (SEM, XRD, hardness)	–	–	–	–	–	–	–	●
Embedded control (STM32 / Arduino)	–	●	–	–	–	–	●	–
Motors & drivers (stepper, servo, TMC2209)	–	●	–	–	–	–	●	–
Sensors & closed-loop control	–	●	–	–	–	–	–	–
Power electronics & wiring	–	●	–	–	–	–	–	–
Hydraulics & PLC control	–	–	–	●	–	–	–	–
Injection molding & tooling	–	–	–	–	●	–	–	–
CNC programming (G-code) & metrology	–	–	–	–	–	●	–	–
Calibration & root-cause analysis	●	●	●	–	–	–	–	–
Lean / 5S / Kaizen process improvement	●	–	–	–	–	–	–	–
Numerical modeling (Python, MD)	–	–	–	–	–	–	●	●
Technical training & SOPs	●	–	–	–	–	–	–	●

● applied in this project · NYU = Research & Course Assistant roles



Vedant Nadoda

Mechanical Engineer · M.S. New York University, December 2026

Seeking full-time roles in mechanical, additive manufacturing, manufacturing and applications engineering, product development, R&D / prototyping, and mechatronics.

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DESIGN · BUILD · TEST · ANALYZE · TROUBLESHOOT · IMPROVE